Landau-Type Theory of Structural Selection in Nonlinear Field Systems

Entropy–Information Scaling and the Critical Coherence Index

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Abstract

We propose a Landau-type theoretical framework for structural selection in nonlinear field systems, building on the Critical Coherence Index (CCI) introduced in the companion papers [3, 5].

The central idea is that dynamical regimes are governed by a competition between coarse-grained histogram activity (a proxy for dynamical reorganisation) and structural information (spatial coherence). Numerical simulations of a one-dimensional ϕ^4 field theory show that the CCI is consistent with the empirical scaling

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \qquad \alpha \approx 2.5, \quad r_s = 0.734, \quad p \sim 10^{-21}.$$

We interpret this scaling within a Landau-type structural free energy, in which the CCI plays the role of a control parameter μ driving a transition between ordered, critical, and chaotic regimes. The structural order parameter is identified with the nearest-neighbour mutual information $\mathcal{O} \sim I_{\mathrm{nn}}$.

This framework is presented as a theoretical proposal and interpretive lens, not as a derived result. The Landau coefficients are not derived from first principles and the connection to the empirical scaling remains an open theoretical problem.

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1 Introduction

Nonlinear field theories admit a wide range of mathematically valid solutions, yet only a subset appears as persistent physical structures. Previous work in this series [1, 2] proposed a structural selection principle based on a MaxEnt Boltzmann weight over the solution manifold, and introduced the Critical Coherence Index (CCI) as a dimensionless instability ratio analogous to Reynolds-type numbers.

The empirical finding of [5] — that the CCI scales approximately as $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ with $\alpha \approx 2.5$ — suggests a deeper structure. In Landau's theory of phase transitions, an order parameter and a control parameter together determine the structure of the free energy and the location of phase boundaries.

The present paper asks:

> *Can the CCI and the structural free energy be cast in the form of a Landau theory, with the mutual information as order parameter and the CCI as control parameter?*

We emphasise from the outset that the answer is *a theoretical proposal*, not a proof. The Landau framework is introduced as an interpretive lens that organises the empirical findings of the companion papers into a coherent theoretical picture.

2 Model

We consider the one-dimensional ϕ^4 scalar field:

$$\ddot{\phi} = \partial_x^2\phi - V'(\phi), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \tag{1}$$

on a periodic lattice of $N = 128$ sites ($\Delta x = 1$), integrated with fourth-order Runge–Kutta ($\Delta t = 0.02$, $T_{\max} = 24$, seed $= 42$). Initial conditions are random mixtures of vacuum, kink, localised, and noise components.

3 Structural Diagnostics

The Critical Coherence Index is defined as [3]:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}\,(1 + \kappa\, U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}} + \varepsilon}. \tag{2}$$

In the numerical implementation used here, the diagnostics reduce to:

$$\Gamma_{\mathrm{inst}} = \frac{\langle \pi^2 \rangle}{\Lambda_\pi + \langle \pi^2 \rangle}, \tag{3}$$

$$\Gamma_{\mathrm{prod}} = \frac{\langle \pi^2 \rangle + \alpha_g \langle |\nabla\phi|^2 \rangle}{S_0 + \langle \pi^2 \rangle + \alpha_g \langle |\nabla\phi|^2 \rangle}, \tag{4}$$

$$\Gamma_{\mathrm{coh}} = \exp\!\big(-\langle |R[\phi]| \rangle\big), \qquad R[\phi] = \partial_x^2\phi - V'(\phi), \tag{5}$$

$$\Gamma_{\mathrm{corr}} = \frac{I_{\mathrm{nn}}}{I_0 + I_{\mathrm{nn}}}, \tag{6}$$

where $\langle \cdot \rangle$ denotes spatial averaging.

4 Entropy and Information

The coarse-grained histogram entropy is:

$$S_{\mathrm{cg}} = -\sum_k p_k \ln p_k, \tag{7}$$

with histogram probabilities p_k of field values. The positive part of its rate of change serves as a dynamical activity proxy:

$$\dot{S}_{\mathrm{cg}}^{(+)} = \max\left(0,\ \frac{dS_{\mathrm{cg}}}{dt}\right). \tag{8}$$

We stress again that this is *not* a thermodynamic entropy production rate. The ϕ^4 system (1) is Hamiltonian; by the Liouville theorem, true phase-space entropy is conserved. $\dot{S}_{\mathrm{cg}}^{(+)}$ measures only how rapidly the field value distribution rearranges.

Structural information is:

$$I_{\mathrm{nn}} = I(\phi_i : \phi_{i+1}), \tag{9}$$

the nearest-neighbour mutual information.

5 Empirical Scaling

The companion study [5] found that the CCI is consistent with the power-law scaling:

$$\mathcal{C}_{\mathrm{CCI}} \sim R_\alpha \equiv \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \tag{10}$$

with optimal exponent $\alpha \approx 2.5$:

$$r_s = 0.734, \quad p \sim 10^{-21} \quad \text{(Spearman)}, \tag{11}$$

$$r = 0.427 \quad \text{(Pearson)}. \tag{12}$$

The Spearman–Pearson discrepancy indicates a monotonic but strongly nonlinear relationship.

6 Landau-Type Theory

6.1 Order Parameter and Control Parameter

We identify the structural order parameter with the nearest-neighbour mutual information:

$$\mathcal{O} \equiv I_{\mathrm{nn}}. \tag{13}$$

High $\mathcal{O}$ corresponds to spatially ordered, coherent configurations (kinks, domain walls); low $\mathcal{O}$ corresponds to disordered, chaotic states.

We identify the control parameter with the CCI:

$$\mu \equiv \mathcal{C}_{\mathrm{CCI}}. \tag{14}$$

6.2 Structural Free Energy

By analogy with Landau theory, we propose the following effective structural free energy:

$$\boxed{\mathcal{F}_{\mathrm{struct}}[\mathcal{O}] = a(\mu)\,\mathcal{O}^2 + b\,\mathcal{O}^4 + \gamma\,(\nabla\mathcal{O})^2} \tag{15}$$

with Landau coefficient:

$$a(\mu) = a_0\,(\mu - \mu_c), \qquad \mu_c = \text{critical CCI value}. \tag{16}$$

The three regimes follow from the standard Landau analysis:

$$\mu < \mu_c \quad \Rightarrow \quad a(\mu) < 0: \text{ non-zero equilibrium } \mathcal{O} \text{ (ordered regime)}, \tag{17}$$

$$\mu = \mu_c \quad \Rightarrow \quad a(\mu) = 0: \text{ critical point}, \tag{18}$$

$$\mu > \mu_c \quad \Rightarrow \quad a(\mu) > 0: \text{ zero equilibrium } \mathcal{O} \text{ (chaotic regime)}. \tag{19}$$

6.3 Connection to the Empirical Scaling

Using the empirical relation (10), the control parameter becomes:

$$\mu \approx \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{2.5}}, \tag{20}$$

so the Landau coefficient (16) is directly linked to the ratio of dynamical activity to structural information:

$$a \approx a_0 \left(\frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{2.5}} - \mu_c \right). \tag{21}$$

This provides a physical picture for the structural transition: the system moves from the ordered to the chaotic phase when dynamical reorganisation overtakes structural coherence by a factor exceeding $\mu_c \cdot I_{\text{nn}}^{2.5}$.

6.4 Status of the Proposal

We emphasise that the Landau framework is here a *theoretical proposal*, not a derived result. Specifically:

1. The Landau coefficients a_0, b, γ, and the critical value μ_c are not derived from the microscopic field theory.
2. The identification $\mathcal{O} \sim I_{\text{nn}}$ is motivated by the empirical findings but not proven.
3. The connection between the scaling exponent $\alpha \approx 2.5$ and the Landau theory remains an open theoretical problem.

7 Discussion

The Landau framework proposed here organises the numerical findings of the companion papers into a coherent theoretical picture, with three advantages:

Unification. The structural free energy of [3], the entropy-information hypothesis of [4], and the empirical scaling of [5] are all consistent with a single Landau-type description.

Prediction. The Landau picture predicts a second-order-like structural transition at $\mu = \mu_c$, with the order parameter I_{nn} growing continuously from zero below the transition. This is testable: a finer sweep of initial conditions near the critical regime should reveal a continuous growth of I_{nn}.

Extension. The Landau form (15) naturally extends to include spatial gradients (γ-term), suggesting a Ginzburg–Landau-type theory for spatially inhomogeneous structural transitions.

8 Conclusion

We have proposed a Landau-type theoretical framework for structural selection in nonlinear field systems. The nearest-neighbour mutual information I_{nn} plays the role of the structural order parameter, and the CCI plays the role of the control parameter.

The empirical scaling relation $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{2.5}$ links the Landau coefficient directly to the ratio of dynamical activity and structural information.

This establishes a conceptual bridge between non-equilibrium dynamical activity, information-theoretic structural measures, and the Landau theory of structural phase transitions.

The derivation of the Landau coefficients from first principles, the theoretical explanation of the scaling exponent $\alpha \approx 2.5$, and the extension to higher-dimensional and gauge field theories remain open problems for future work.

This result is currently empirical and derived from numerical simulations of a single 1D model. A first-principles derivation of the scaling relation, the Landau coefficients, and the critical value μ_c remains an open problem. Universality across different field theories and dimensions has not yet been established.

References

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